Gluco-X Nudge Effectiveness Analysis

Executive Summary for the Product Team

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AI Disclosure: This report was developed with the assistance of Claude (Anthropic). Claude was used to create the document structure based on the Jupyter notebook used for the analysis, and to generate the key findings table.

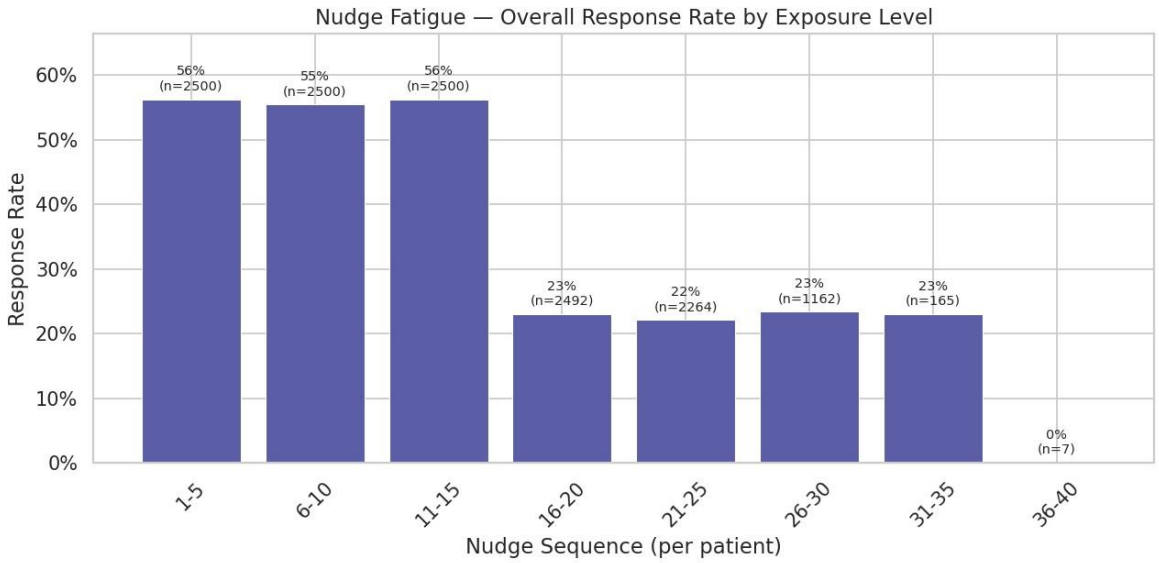
1. Key Findings at a Glance

Metric	Value
Total nudges sent	13,590
Total nudges responded to	5,582
Overall response rate	41.1%
Gentle Reminder response rate	42.4%
Urgent Alert response rate	39.8%
Response rate (nudges 1-15)	~56%
Response rate (nudges 16+)	~23%
Fatigue cliff starts at	Nudge #16

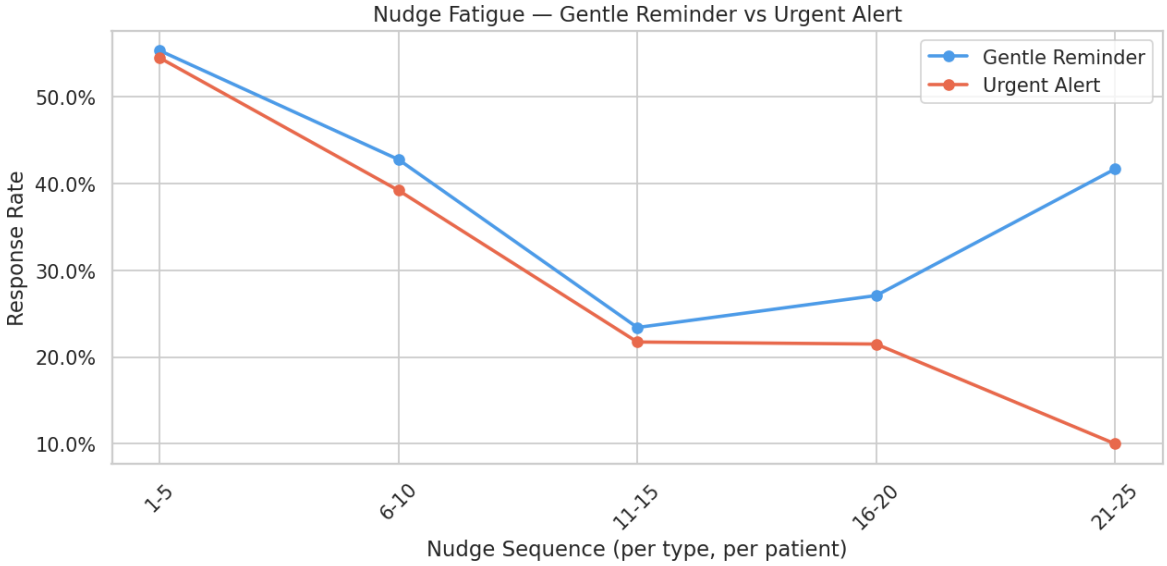
The data confirms the Product Manager hypothesis: nudge fatigue is real and measurable**.** Patients respond to roughly 56% of their first 15 nudges, but response rates collapse to around 23% once cumulative exposure exceeds 15 nudges. This cliff is observed for both nudge types.

2. Nudge Fatigue Analysis

The chart below shows the overall response rate as a function of cumulative nudge exposure. The drop-off between buckets 11-15 and 16-20 is the most pronounced decline in the dataset, representing a nearly 33 percentage point decrease.

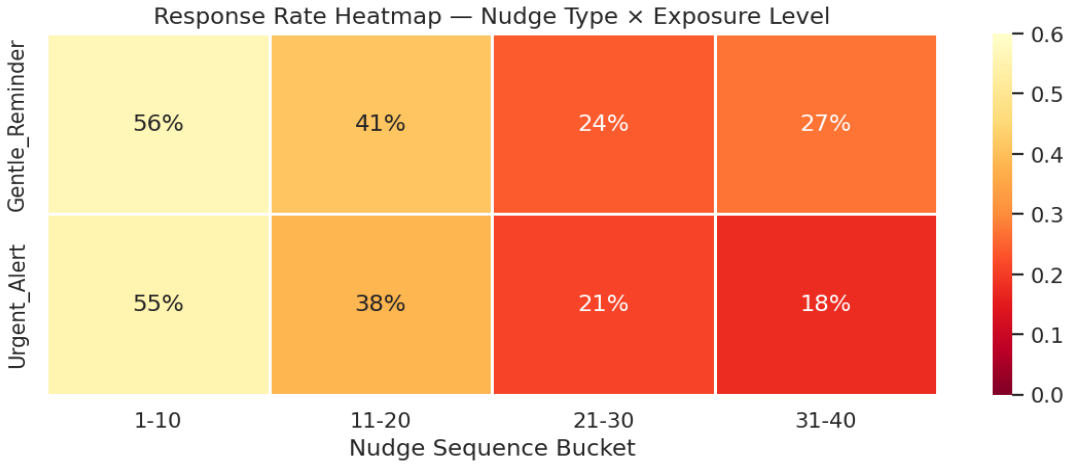


Splitting by nudge type reveals that **both Gentle Reminders and Urgent Alerts decay at similar rates**. The Urgent Alert does not meaningfully outperform the Gentle Reminder at any exposure level, contradicting the assumption that urgent framing sustains engagement longer.



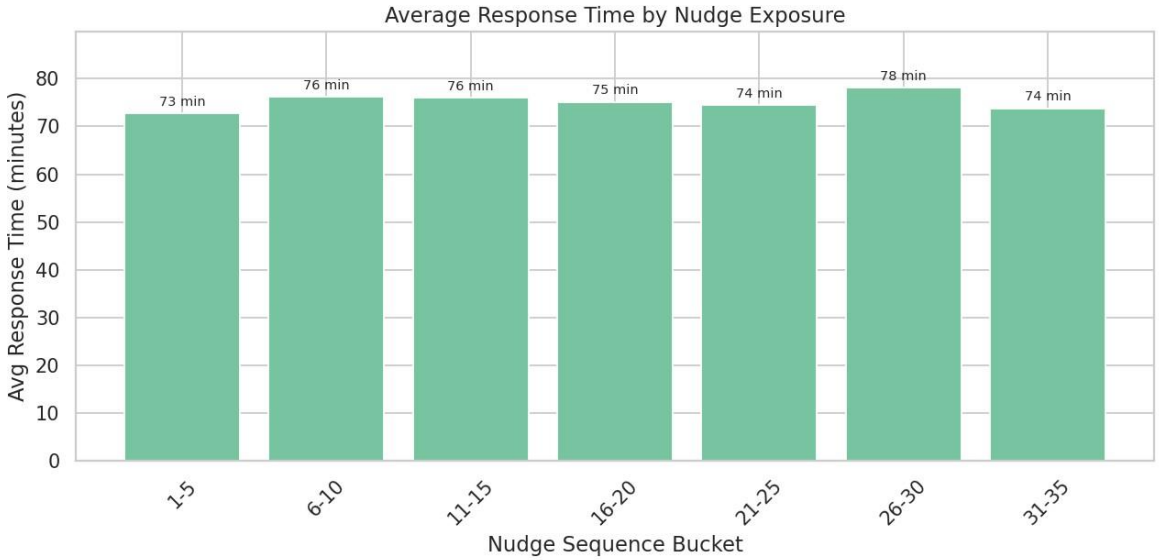
3. Response Rate Heatmap

The heatmap provides a compact view of how response rates vary across nudge type and exposure level. The first 10 nudges for both types cluster around 40-55%. Beyond that, rates fall below 25% regardless of nudge type.



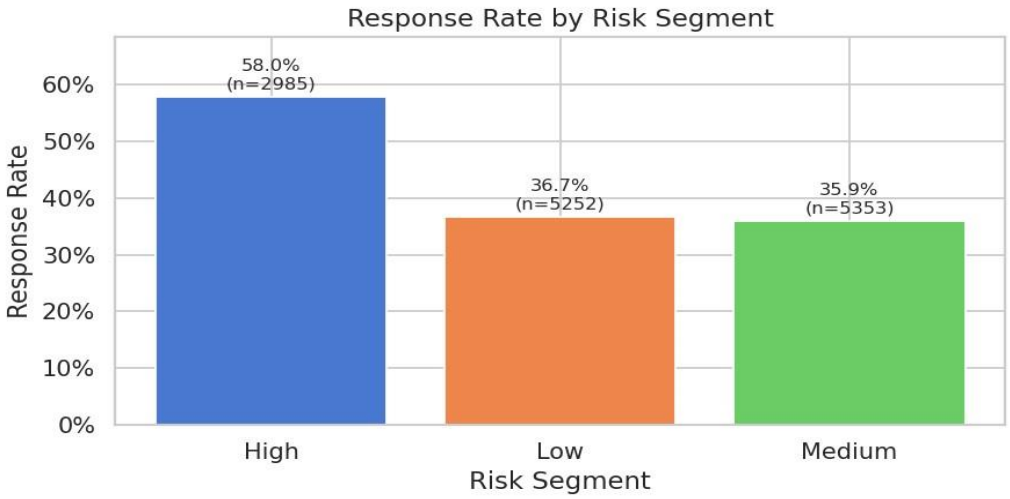
4. Response Time Trend

Among patients who do respond, the average response time remains relatively stable across exposure levels. This suggests that fatigue manifests as complete disengagement rather than slower reactions: patients either respond promptly or ignore the nudge entirely.

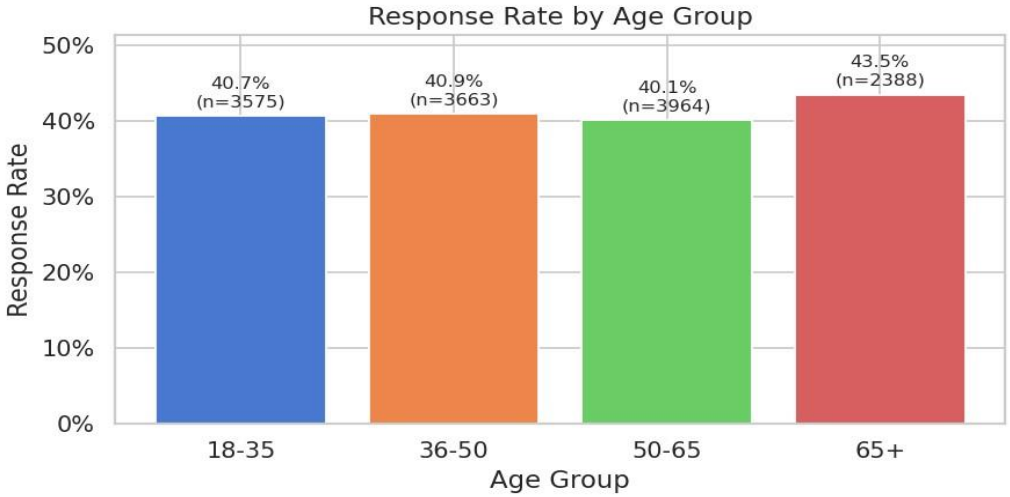


5. Demographic Segmentation

Risk Segment: High-risk patients respond at 58%, nearly 20 percentage points above Low and Medium risk cohorts (~36%). This is clinically intuitive: patients who perceive greater personal health stakes are more motivated to act.



Age Group: Response rates are fairly uniform across age groups (40-43%), with the 65+ cohort showing a marginal advantage. Age alone is not a strong predictor of engagement.



6. Recommendations

- **Implement a nudge frequency cap.** Limit each patient to a maximum of 12-15 nudges before rotating strategy (For example, a cooldown period or a phone call). The data shows diminishing returns beyond this threshold.
- **Retire the two-type A/B framing.** Gentle Reminders and Urgent Alerts show no statistically meaningful difference in response rate or fatigue trajectory. Invest in testing fundamentally different nudge modalities (e.g., personalized insights, gamification, social proof) rather than tone variations.
- **Prioritize the High-risk segment.** High-risk patients are the most responsive cohort. Tailor nudge content and cadence specifically for them to maximize clinical impact. Allocate more frequency 'budget' to high-risk patients while further reducing the frequency for low-risk patients to avoid premature burnout.
- **Track re-engagement metrics.** Extend the pipeline to measure whether patients who go dormant can be recaptured with new nudge strategies after a cooling-off period.

7. Methodology Notes

Attribution logic: A glucose measurement is credited to the **most recent nudge** sent to the same patient within the preceding **4-hour window**. Each nudge can receive at most one response credit (the earliest qualifying measurement). Nudges with no measurement within the window are marked as unresponded.

The fatigue curve bins are computed using the patient-level sequential nudge counter, grouped into buckets of 5. Response rate is defined as (nudges responded) / (nudges sent) within each bucket. Demographic breakdowns use the patient registry joined on patient_id.